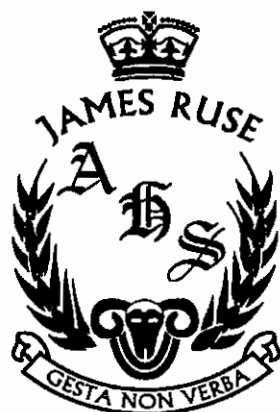


Student Number:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2018

MATHEMATICS EXTENSION 2

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 3 hours.
- Write in black pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 11 - 16, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 100

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 90 Marks

- Attempt Question 11 - 16
- Answer on lined paper provided. Start a new page for each new question.
- Allow about 2 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section A MULTIPLE CHOICE

(Use the Multiple choice answer sheet for questions 1 – 10)

One mark will be awarded for each correct answer.

Question 1

After using the method of slicing to find a volume, the final calculation requires the evaluation of this integral

$$\int_{-8}^8 \sqrt{64 - x^2} \, dx$$

The value of this integral is :

- A) 0 B) 32π C) 64π D) 128π

Question 2

If ω is a complex root of the equation $z^3 = 1$, then the value of

$$\frac{1}{5+6\omega+5\omega^2} - \frac{1}{3+3\omega+2\omega^2} \text{ is}$$

- A) -1 B) $-\frac{1}{2}$ C) $\frac{1}{2}$ D) 1

Question 3

If the polynomial $P(x)$ is divided by $(x^2 + 25)$ then the remainder is $3x + 4$.

What is the remainder when $P(x)$ is divided by $(x + 5i)$?

- A) $15 - 4i$ B) $15 + 4i$ C) $4 - 15i$ D) $4 + 15i$

Question 4

The gradient of the tangent to the curve $x^3 + y^3 - 8xy + 7 = 0$ at the point (1,2) is:

- A) $\frac{4}{13}$ B) Undefined C) $\frac{13}{4}$ D) $\frac{3}{2}$

Question 5

A projectile is fired from the origin at a fixed angle θ with a velocity V . If this velocity is doubled, while the angle of projection remains the same, then the range of the projectile across horizontal ground will:

- A) Remain unchanged.
- B) Increase by a factor of $\sqrt{2}$.
- C) Increase by a factor of 2 .
- D) Increase by a factor of 4.

Question 6

Which of the following loci is identical to the locus $|z| = 2$, $\text{Im}(z) > 0$?

- A) $z\bar{z} = 4$, $\text{Re}(z) > 0$
- B) $\text{Arg}\left(\frac{z-2}{z+2}\right) = \frac{\pi}{2}$
- C) $\text{Re}\left(\frac{z+2}{z-2}\right) = 0$
- D) $\text{Arg}\left(\frac{z+2}{z-2}\right) = \frac{\pi}{2}$

Question 7

What is the solution of the inequality $\frac{x(5-x)}{x-4} \geq -3$?

- A) $2 \leq x < 4$ or $x \geq 6$
- B) $4 < x \leq 5$ or $x \leq 1$
- C) $4 < x \leq 6$ or $x \leq 2$
- D) $1 \leq x < 4$ or $x \geq 5$

Question 8

$f(x)$ is a continuous, differentiable function over $a \leq x \leq b$ and $g(x)$ is a continuous, differentiable function over $c \leq x \leq d$. Which one of the following integrals is always greater than, or equal to, the other choices?

- A) $\int_a^b f(x) dx + \int_c^d g(x) dx$
- B) $\left| \int_a^b f(x) dx \right| + \left| \int_c^d g(x) dx \right|$
- C) $\int_a^b |f(x)| dx + \int_c^d |g(x)| dx$
- D) $\left| \int_a^b f(x) dx + \int_c^d g(x) dx \right|$

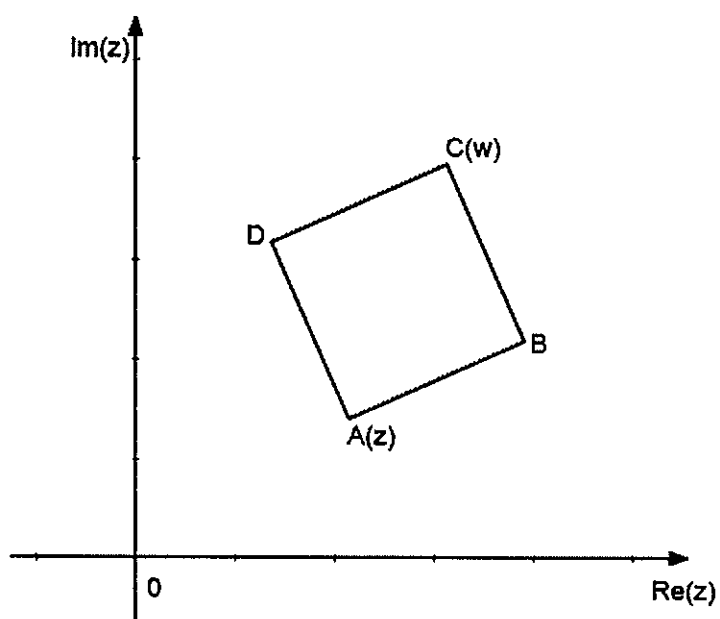
Question 9

A particle is moving along a straight line so that initially its displacement is $x=1$, its velocity $v=2$ and its acceleration is $a=4$.

Which of the following is a possible equation describing the motion of the particle?

- A) $v^2 = 4(x^2 - 2)$ B) $v = 2 + 4\log_e x$
C) $v = 2 \sin(x - 1) + 2$ D) $v = x^2 + 2x + 4$

Question 10



In the diagram, $ABCD$ is a square in the first quadrant as shown. If A represents the complex number z and C represents the complex number w , then B represents which of the following complex numbers?

- A) $\frac{z-w}{2} - \frac{i(z+w)}{2}$ B) $\frac{z-w}{2} + \frac{i(z+w)}{2}$
C) $\frac{z+w}{2} - \frac{i(z-w)}{2}$ D) $\frac{z+w}{2} + \frac{i(z-w)}{2}$

End of Section A

Section B

In this section you should include all necessary reasoning and calculations.

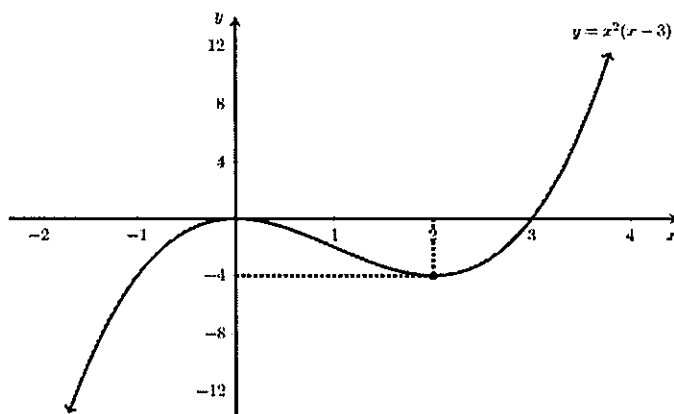
Question 11 (Start each new question on a separate sheet of paper.)

Marks

- (a) By using partial fractions, or otherwise, evaluate $\int_{-2}^{-1} \frac{18dx}{x^2(x+3)}$ **3**
- (b) Evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$ **3**
- (c) (i) Show that $\frac{d}{dx}(x^2 \tan^{-1} x) = 2x \tan^{-1} x + 1 - \frac{1}{1+x^2}$ **2**
- (ii) Hence find the exact value of $\int_0^{\sqrt{3}} x \tan^{-1} x \, dx$ **2**
- (d) $z_1 = -2 + 2i$ and $z_2 = -1 + i\sqrt{3}$ are two complex numbers.
- (i) Express z_1 , z_2 and $\frac{z_1}{z_2}$ in modulus/argument form. **3**
- (ii) Find the smallest positive integer n such that $\frac{z_1^n}{z_2^n}$ is purely imaginary and, for this value of n , write down the value of $\frac{z_1^n}{z_2^n}$ in the form bi where b is a real number. **2**

Question 12 (Start each new question on a separate sheet of paper.)**Marks**

(a)



A sketch of the curve $y = x^2(x - 3)$ is shown above.

On separate sets of axes sketch the graphs of:

(i) $y^2 = x^2(x - 3)$ 2

(ii) $y = \frac{1}{x^2(x-3)}$ 2

(iii) $y = \frac{1}{x^2(|x|-3)}$ 1

(b) Consider the ellipse $4x^2 + 3y^2 = 48$.

(i) Find the eccentricity of the ellipse. 1

(ii) Write down the coordinates of the foci and the equations of the directrices. 2

(iii) Sketch the ellipse, showing the above features and the intercepts on the coordinate axes. 1

(c) (i) Prove that, for all real a and b ,

$$a^2 + b^2 \geq 2ab$$
 1

(ii) Prove that, for any positive, real x and y ,

$$\frac{x}{y} + \frac{y}{x} \geq 2$$
 1

(iii) Prove by induction, or otherwise, that

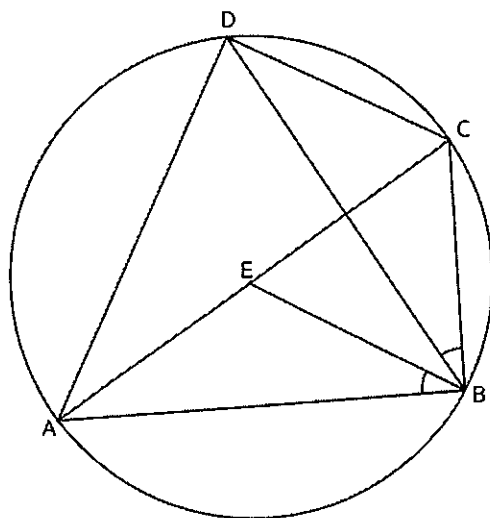
$$(x_1 + x_2 + x_3 + \cdots + x_{n-1} + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_{n-1}} + \frac{1}{x_n} \right) \geq n^2$$

where the x_i are all real and positive.

4

Question 13

(Start each new question on a separate sheet of paper.)

Marks

- (a) The figure shows a cyclic quadrilateral $ABCD$ with diagonals AC and BD . E is a point on AC such that angle ABE equals angle DBC .

- (i) Prove that triangle ABE is similar to triangle DBC . 2
- (ii) Prove that triangle ABD is similar to triangle EBC . 2
- (iii) Hence prove Ptolemy's Theorem, which, in this context, states

$$AB \times CD + AD \times BC = AC \times BD$$
 3

- (b) Sketch the curve $y = \frac{(x-2)(x+1)}{5-x}$ over the domain $\{x: -5 < x < 15\}$ showing the axis intercepts and the relationship of the curve to its oblique asymptote. (You are given that there are turning points near $x = 0.75$ and $x = 9.25$ but you need not evaluate their values exactly.) 3

- (c) A jar contains w white and r red jelly beans. Three jellybeans are taken at random from the jar and eaten.

- (i) Write down an expression, in terms of w and r , for the probability that the three jellybeans were all white. (Leave in factorised form.) 1

It was observed that, if the jar had initially contained $(w + 1)$ white and r red jellybeans, then the probability that the 3 eaten jellybeans were white would have been double that in part (i).

- (ii) Show that $r = \frac{(w-2)(w+1)}{5-w}$ 2
- (iii) Using part (b) or otherwise, determine all possible values of w and r . 2

Question 14

(Start each new question on a separate sheet of paper.)

Marks

- (a) Consider the polynomial $p(x) = ax^4 + bx^3 + cx^2 + dx + f$ where a, b, c, d and f are integers. Suppose α is a non-zero integer such that $p(\alpha) = 0$.

(i) Prove that α divides f .

1

(ii) Show that the polynomial $q(x) = 4x^4 - x^3 + 3x^2 + 2x - 3$ does not have an integer zero.

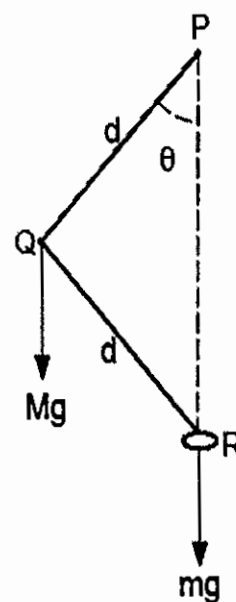
2

- (b) Two light rigid rods PQ and QR , each of length d metres, are smoothly jointed at Q and the rod PQ is smoothly jointed at the fixed point P . The joint at Q has a mass of M kg attached.

A small ring of mass m kg is smoothly jointed to QR at R and can slide on a smooth vertical rod which also passes through P .

The system rotates about the vertical causing the ring to rise up the vertical rod.

In equilibrium the system is rotating at ω rads/sec and the rods form an angle θ with the vertical (as shown).



- (i) Copy the diagram and, by resolving forces at points Q and R , show that $M\omega^2 d = (2mg + Mg)\sec\theta$

4

- (ii) Now suppose that $M = 4m$ and that a safety switch operates when $\theta = \pi/3$. Show that this occurs when $\omega = \sqrt{\frac{3g}{d}}$

1

- (c) The complex number z satisfies the condition $|z - 8| = 2 \operatorname{Re}(z - 2)$.

(i) Write down the equation of the locus in its more usual Cartesian form, noting any restrictions that apply.

3

(ii) Sketch the locus on an Argand diagram and find the possible values of $\operatorname{Arg} z$.

3

(iii) Write down the value of $|z + 8| - |z - 8|$ for any point on the curve.

1

Question 15

(Start each new question on a separate sheet of paper.)

Marks

- (a) The area bounded by the parabola $y = x(3 - x)$ and the x axis is rotated about the line $x = 4$. Use the method of cylindrical shells to find the volume of the solid formed. 4
- (b) Let α , β and γ be the roots of the equation $x^3 - 3x^2 + 5 = 0$.
- (i) Find a cubic polynomial equation with integer coefficients whose roots are α^2 , β^2 and γ^2 . 2
- (ii) Find the value of $\alpha^4 + \beta^4 + \gamma^4$. 2
- (c) $P(3p, 3/p)$ and $Q(3q, 3/q)$ are two points on the rectangular hyperbola $xy = 9$. The equation of the chord PQ is $pqy + x = 3(p + q)$ (DO NOT PROVE THIS)
- The chord PQ meets the x and y axes at G and H respectively.
- (i) Find the coordinates of M , the midpoint of PQ . 1
- (ii) Show that $PH = QG$ 2
- (iii) If the chord PQ always passes through the point $(0, 2)$, find the locus of M , proving any restrictions. 4

Question 16 (Start each new question on a separate sheet of paper.)**Marks**

- (a) A particle P , of unit mass, is thrown vertically downwards in a resistive medium where the resistive force is proportional to speed. Thus, taking downwards as positive, the equation of motion is $\ddot{x} = g - kv$ where $k > 0$.

The particle is thrown, with an initial speed U , from a point T which is d units above a fixed point O which should be taken as the origin.

So the initial conditions are : when $t = 0$, $v = U$ and $x = -d$

- (i) Show that , during the subsequent motion of P ,

$$v = \frac{g}{k} - \left(\frac{g - kU}{k} \right) e^{-kt}$$

(You may assume that $U < \frac{g}{k}$)

2

- (ii) Integrate again to show that

$$x = \frac{gt - kd}{k} + \left(\frac{g - kU}{k^2} \right) (e^{-kt} - 1).$$

2

- (iii) An identical particle Q is dropped from rest at O at the same instant that P is thrown down. Use the above results to write down expressions for v and x as functions of time for Q .

2

- (iv) The particles P and Q collide. Find when the collision occurs and the difference in speeds as the particles collide.

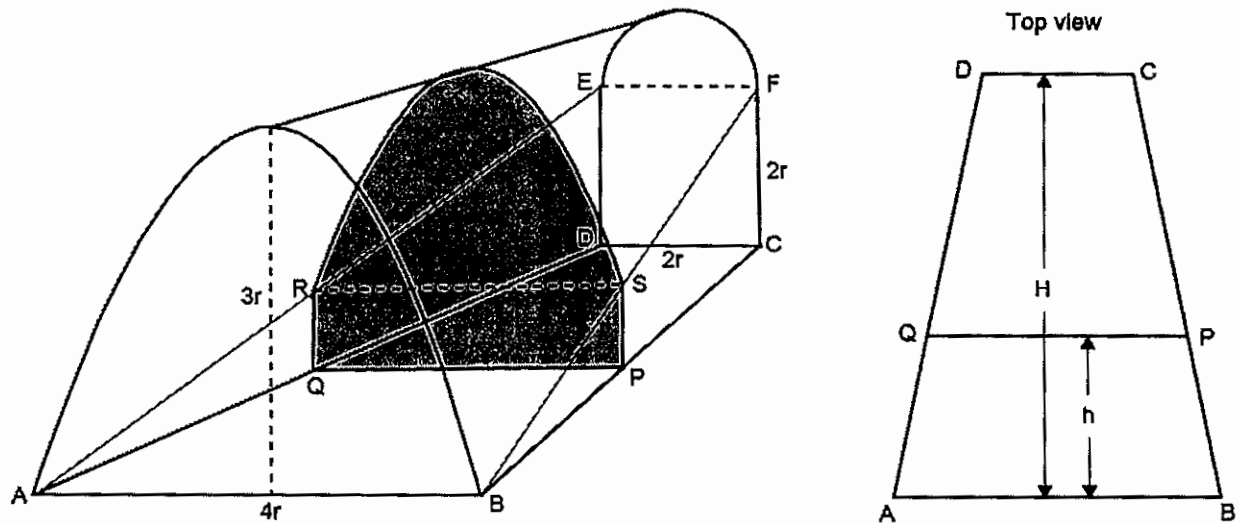
3

Question 16 is continued on the next page...

Question 16 (continued)

Marks

(b)



A tunnel has an isosceles trapezium $ABCD$ as its base with AB of side length $4r$ and CD of side length $2r$. The tunnel is of length H and the height along the centre line is constant at $3r$. The cross section perpendicular to the floor at AB is a semi-ellipse with AB as the minor axis and with a semi-major axis of length $3r$. The cross section perpendicular to CD is a square $CDEF$ surmounted by a semicircle of radius r .

A typical internal cross-section (as shown) comprises a rectangle surmounted by a semi-ellipse. AED and BCF are thence plane triangles.

(You may assume that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab square units.)

- (i) Show that the tunnel cross-sectional area at a perpendicular distance h from AB is given by :

$$\frac{r^2}{2H^2} (2H - h)(3\pi H + (8 - 2\pi)h) \quad 4$$

- (ii) Hence find the volume of the tunnel. 2

END OF EXAMINATION

MULTIPLE CHOICE

1. B 2. A 3. C 4. C 5. D 6. B 7. C 8. C 9. C 10. D

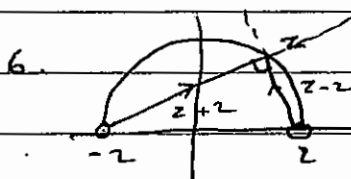
1. Semicircle rad = 8. $\pi 8^2/2$

2. $1 + w + w^2 = 0$ $\frac{1}{w} = \frac{1}{-w^2} = \frac{1}{w} + \frac{1}{w^2} = \frac{w + w^2}{w^3} = -1$

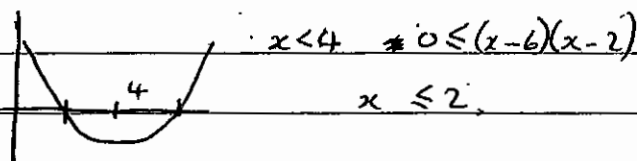
3. $P(x) = Q(x)(x+5i)(x-5i) + 3x + 4$
 $P(-5i) = -15i + 4$

4. $3x^2 + 3y^2 y' - 8y - 8xy' = 0$ $y' = \frac{8y - 3x^2}{3y^2 - 8} = \frac{16 - 3}{12 - 8}$

5. Range = $\frac{v^2 \sin 2\theta}{g}$



7. $x > 4$ $x(5-x) \geq -3(x-4)$
 $0 \geq x^2 - 8x + 12$
 $0 \geq (x-6)(x-2)$
 $4 < x \leq 6$



8. C

9. a) $v^2 < 0$ b) $\int \frac{v dv}{dx} = \frac{(2+4 \ln x) 4}{x} = 8 \ln x = 1 \times$ d) $v(1) \neq 1 \therefore c \checkmark$

10. $OB = OA + AB$

$$= OA + AC \frac{1}{\sqrt{2}} \cos(-45^\circ)$$

$$= z + (w-z) \frac{1}{\sqrt{2}} \left(\frac{1-i}{\sqrt{2}} \right) = z + (w-z) \left(\frac{1-i}{2} \right)$$

$$= \cancel{z} + w \left(\frac{1-i}{2} \right) + z \left(1 - \frac{1-i}{2} \right) = \frac{(z+w)}{2} + \frac{(z-w)i}{2}$$

MATHEMATICS Extension 2: Question 11

Suggested Solutions	Marks	Marker's Comments
(a) $\frac{18}{x^2(x+3)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{x+3}$ $A(x+3) + Bx(x+3) + Cx^2 = 18$ $x = -3 \quad C = 2$ $x = 0 \quad 3A = 18 \quad A = 6$ $x = 1 \quad 4A + 4B + C = 18$ $24 + 4B + 2 = 18$ $4B = -8$ $B = -2$ $\int_{-2}^{-1} \left(\frac{6}{x^2} - \frac{2}{x} + \frac{2}{x+3} \right) dx$ $= \left[-\frac{6}{x} - 2 \ln x + 2 \ln x+3 \right]_{-2}^{-1}$ OR $-\ln x^2$ $= (6 + 2 \ln 2) - (3 - 2 \ln 2 + 0)$ $= 3 + 4 \ln 2$ b) $\int_0^{\pi/4} \sec^4 x dx = \int_0^{\pi/4} \sec^2 x \sec^2 x dx$ $= \int_0^{\pi/4} (1 + \tan^2 x) \sec^2 x dx$	1 1 1	Other methods also valid $\frac{A}{x^2} + \frac{B}{x+3}$ is not correct NB $\int \frac{-6}{x^2}$ $\neq -\frac{1}{3x^3}$ x critical as $\ln(-2)$ not defined careless errors with negative

MATHEMATICS Extension 2 : Question 1

Suggested Solutions	Marks	Marker's Comments
let $u = \tan x$ $\frac{du}{dx} = \sec^2 x$ $x = \frac{\pi}{4}$ $u = 1$ $x = 0$ $u = 0$ $\int_0^1 (1+u^2) du$ $= \left[u + \frac{u^3}{3} \right]_0^1$ $= \frac{4}{3}$	1 1	students made this more difficult by using integration by parts.

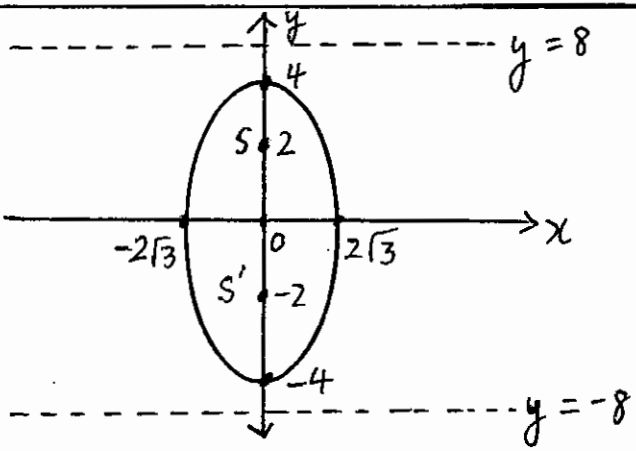
MATHEMATICS Ext 2: Question 11

Suggested Solutions	Marks	Marker's Comments
(c) (i) $\frac{d}{dx}(x^2 \tan^{-1} x) = 2x \tan^{-1} x + x^2 \times \frac{1}{1+x^2}$ (product rule) (ii) $\int_0^{\sqrt{3}} x \tan^{-1} x = \frac{1}{2} [x^2 \tan^{-1} x]_0^{\sqrt{3}}$ $- \frac{1}{2} \int_0^{\sqrt{3}} 1 dx + \frac{1}{2} \int_0^{\sqrt{3}} \frac{dx}{1+x^2}$ $= \frac{1}{2} [x^2 \tan^{-1} x - x + \tan^{-1} x]_0^{\sqrt{3}}$ $= \frac{1}{2} \left[3 \times \frac{\pi}{3} - \sqrt{3} + \frac{\pi}{3} - 0 \right]$ $= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$	1 1	
(d) $Z_1 = 2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$ $Z_2 = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$ $\frac{Z_1}{Z_2} = \sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4} - \frac{2\pi}{3} \right) \text{ (de Moivre's)}$ $= \sqrt{2} \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}$	1 1 1	Students who wrote cis lost a mark. A number of students found point in wrong quadrant

MATHEMATICS Extension 2 : Question 5

Suggested Solutions	Marks	Marker's Comments
(ii) $\left(\frac{z_1}{z_2}\right)^n = (\sqrt{2})^n \left(\cos \frac{n\pi}{12} + i \sin \frac{n\pi}{12}\right)$ (De Moivre's) Purely imaginary when $\frac{n\pi}{12} = \frac{\pi}{2}$ ie $n = 6$ need $\therefore \left(\frac{z_1}{z_2}\right)^6 = (\sqrt{2})^6 i \sin \frac{\pi}{2}$ $= \underline{8i}$ $b = 8$	1 1	needed to show working

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JRAHS : TRIALS 2018 MATHEMATICS Extension 2: Question 12		
Suggested Solutions	Marks	Marker's Comments
b) (i) $4x^2 + 3y^2 = 48$ $\Rightarrow \frac{x^2}{12} + \frac{y^2}{16} = 1$ which is of the form $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ with $a > b$ [Vertical ellipse] Hence $b^2 = a^2(1 - e^2)$ $12 = 16(1 - e^2)$ $\therefore e = \frac{1}{2}$, $0 < e < 1$	1	Note for transverse ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $a > b$ and $b^2 = a^2(1 - e^2)$ For correct $e > 0$
(ii) Foci at $(0, \pm ae)$ $(0, \pm 4 \times \frac{1}{2})$ $(0, \pm 2)$ directrix at $y = \pm \frac{a}{e}$ $= \pm \frac{4}{\frac{1}{2}}$ $= \pm 8$	1 1	Note for horizontal ellipse, Foci at $(\pm ae, 0)$ For Foci Note for horizontal ellipse, directrix at $x = \pm \frac{a}{e}$ For directrix
(iii) 	1	For Foci, directrix and intercepts

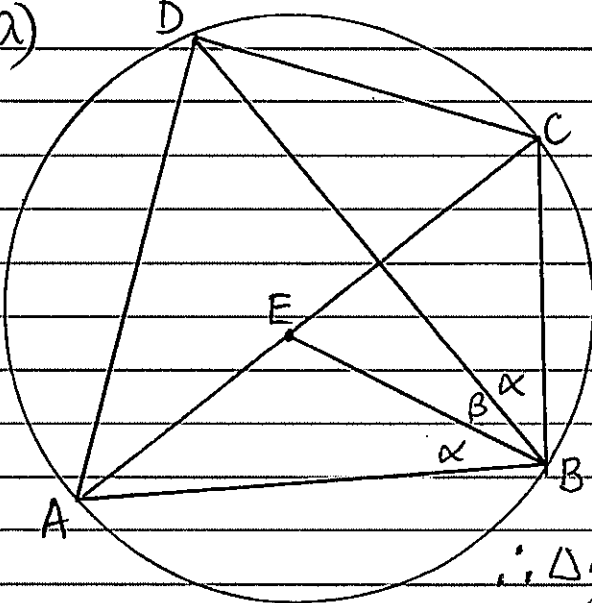
JRAHS : TRIALS 2018 MATHEMATICS Extension 2: Question 12

Suggested Solutions	Marks	Marker's Comments
c) (i) For $a, b \in \mathbb{R}$, $(a-b)^2 \geq 0$ $\Rightarrow a^2 - 2ab + b^2 \geq 0$ $\therefore a^2 + b^2 \geq 2ab$ <u>Method II</u>: Consider $a^2 + b^2 - 2ab$ <u>Method III</u>: Assume $a^2 + b^2 \leq 2ab$ and arrive at a contradiction	1	For correct proof
(ii) Let $a = \sqrt{x/y}$ and $b = \sqrt{y/x}$ where $x, y > 0$ Since $a^2 + b^2 \geq 2ab$, from (i) then $(\sqrt{x/y})^2 + (\sqrt{y/x})^2 \geq 2\sqrt{x/y} \cdot \sqrt{y/x}$ $\Rightarrow \frac{x}{y} + \frac{y}{x} \geq 2$ <u>Method II</u>: From (i) $x^2 + y^2 \geq 2xy$ $\Rightarrow \frac{x^2 + y^2}{xy} \geq 2$ if $x, y > 0$ and $xy > 0$ $\Rightarrow \frac{x^2}{xy} + \frac{y^2}{xy} \geq 2$ i.e. $\frac{x}{y} + \frac{y}{x} \geq 2$ <u>Method III</u>: Consider $\frac{x}{y} + \frac{y}{x} - 2$ and proceed as usual	1	For correct proof

JRAHS : TRIALS : 2018 MATHEMATICS Extension 2: Question 12

Suggested Solutions	Marks	Marker's Comments
(iii) To prove $(x_1 + x_2 + \dots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) \geq n^2$ <u>Proof</u>: Base case: Prove true for $n=1$ LHS = $x_1 \times \frac{1}{x_1} = 1$ RHS = $1^2 = 1$ $\therefore$ LHS $\geq$ RHS $\therefore$ true for $n=1$ <u>Assume true for $n=k \in \mathbb{Z}^+$</u> i.e. $(x_1 + x_2 + \dots + x_k) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k} \right) \geq k^2$ <u>Prove statement true for $n=k+1$</u> i.e. to show/prove $[(x_1 + \dots + x_k) + x_{k+1}] \left[\left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + \frac{1}{x_{k+1}} \right] \geq (k+1)^2$ Now LHS = $[A + x_{k+1}] \left[B + \frac{1}{x_{k+1}} \right]$ where $A = (x_1 + x_2 + \dots + x_k)$ $B = \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k} \right)$ $= AB + A \cdot \frac{1}{x_{k+1}} + B \cdot x_{k+1} + \frac{x_{k+1}}{x_{k+1}}$ $\geq k^2 + \left[\left(\frac{x_1}{x_{k+1}} + \frac{x_{k+1}}{x_1} \right) + \left(\frac{x_2}{x_{k+1}} + \frac{x_{k+1}}{x_2} \right) + \dots + \left(\frac{x_k}{x_{k+1}} + \frac{x_{k+1}}{x_k} \right) \right] + 1$ $\geq k^2 + [2 + 2 + \dots + 2] + 1$ $= k^2 + 2k + 1$ $= (k+1)^2 = \text{RHS}$ Hence by PMI, the statement is true for all $n \in \mathbb{Z}^+$	n^2 1 1 1 1	for $n \geq [?]$ $x_i \in \mathbb{R}^+$ For Base case For Assumption For using $A \cdot B \geq k^2$ For use of part(ii) $A \cdot B \geq k^2$ By assumption by part (ii)

13

Suggested Solutions		Marks	Marker's Comments
a)	 i) In $\triangle ABE$ & $\triangle DBC$ $\angle ABE = \angle CBD$ (given) $\angle BAE = \angle BDC$ (angles at the circumference standing on arc BC are equal) $\therefore \triangle ABE \parallel \triangle DBC$ (equiangular)	1 1	2 pairs of equal angles Need "at the circumference" Please check your wording.
ii)	In $\triangle ABD$ & $\triangle ECB$ $\angle ABD = \angle ECB$ (Both are sum of given equal angles and $\angle EBD$) $\angle ADB = \angle ECB$ (angles at the circumference standing on the same arc AB are equal) $\therefore \triangle ABD \parallel \triangle ECB$ (equiangular)	1 1	2 pairs of equal angles
iii)	$\frac{AB}{DB} = \frac{AE}{EC}$ (corresponding sides of similar triangles ABE and DBC are in the same ratio) Similarly in $\triangle ABD$ and $\triangle ECB$ $\frac{BC}{BD} = \frac{EC}{AD}$ $AB \times CD = AE \times BD$ ① and $AD \times BC = EC \times BD$ ② ① + ② $AB \times CD + AD \times BC = BD (AE + EC)$ as AEC is a straight line	1 1	Check wording of reason combining

MATHEMATICS Extension 2: Question 13

Suggested Solutions

Marks

Marker's Comments

$$b) y = \frac{(x-2)(x+1)}{5-x} = \frac{x^2 - x - 2}{5-x}$$

$$\begin{array}{r} -x-4 \\ x-5 \overline{) -x^2+x+2} \end{array}$$

$$\begin{array}{r} -x^2+5x \\ \underline{-x^2+x+2} \\ 4x-3 \end{array}$$

$$\begin{array}{r} 4x-20 \\ \underline{4x-3} \\ -17 \end{array}$$

$$\therefore y = -x-4 - \frac{17}{x-5}$$

$\therefore$ Asymptotes are $y = -x-4$, $x=5$

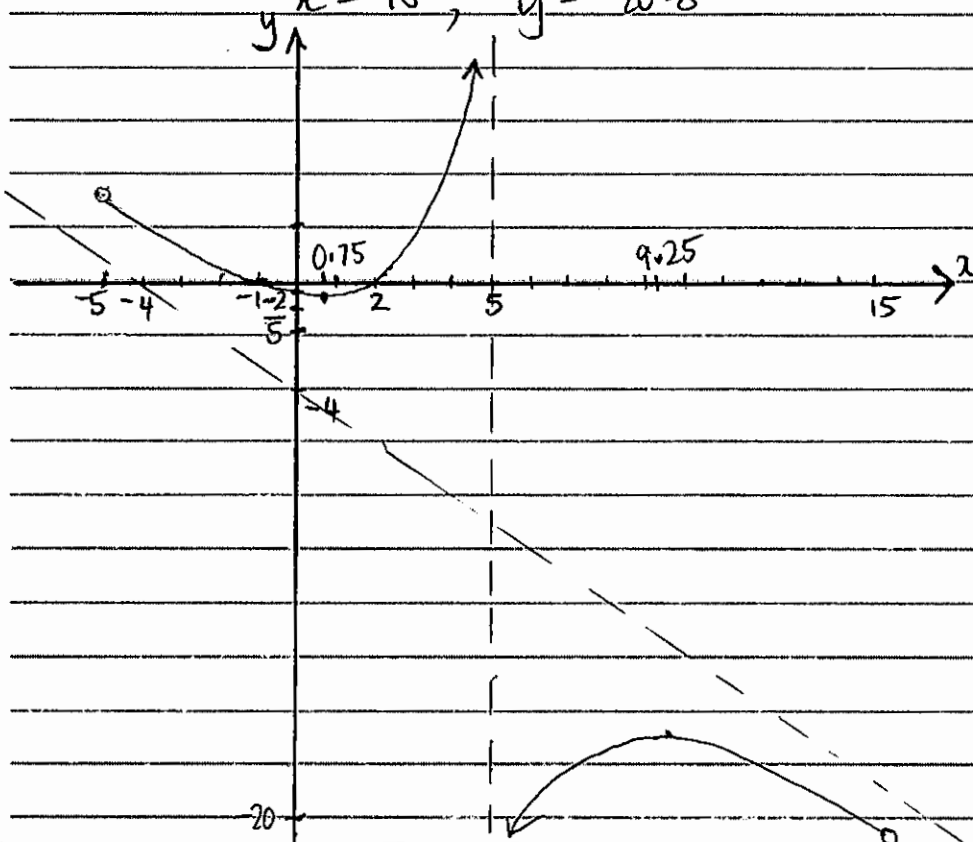
x -intercepts are 2 and -1

y -intercept is $-2/5$

When $x=0.75$, $y=0.5$ } turning
 $x=9.25$, $y=-17.5$ } points

endpoints: $x=-5$, $y=2.8$

$x=15$, $y=-20.8$



- 1 Asymptotes
- 1 x -intercepts
- 1 y -intercept
- 1 extrema
- 1 x -values of TPs

must have 5
out of these
6.

- 1 shape

MATHEMATICS Extension 2: Question...13...

Suggested Solutions

Marks

Marker's Comments

c) i) $\left(\frac{w}{w+r}\right)\left(\frac{w-1}{w+r-1}\right)\left(\frac{w-2}{w+r-2}\right)$ OR, $\frac{{}^wC_3}{{}^{w+r}C_3}$

1

ii) $\left(\frac{w+1}{w+r+1}\right)\left(\frac{w}{w+r}\right)\left(\frac{w-1}{w+r-1}\right) = 2\left(\frac{w}{w+r}\right)\left(\frac{w-1}{w+r-1}\right)\left(\frac{w-2}{w+r-2}\right)$

1

correct equation from your (i)

$$\therefore \frac{w+1}{w+r+1} = \frac{2(w-2)}{w+r-2}$$

$$(w+1)(w+r-2) = 2(w-2)(w+r+1)$$

$$w^2 + rw - 2w + w + r - 2 = 2w^2 + 2wr + 2w - 4w - 4r - 4$$

$$r(-w+5) = w^2 - w - 2$$

$$r = \frac{w^2 - w - 2}{-w+5}$$

$$= \frac{(w-2)(w+1)}{5-w}$$

1

simplification

iii) From graph in b)
 $r, w > 0$ and integers

1

statement including "integers"

$\therefore w = 3$ or 4 only

when $w = 3, r = 2$

$w = 4, r = 10$

1

final answers

Note: $r \neq 0$ as this would give
 $w = 2$, but 3 jellybeans
 are taken $\therefore w > 2$.

Only one mark given if only one pair found

MATHEMATICS Extension 2: Question...14...

Suggested Solutions

Marks

Marker's Comments

a) i) $p(x) = ax^4 + bx^3 + cx^2 + dx + f$

$p(\alpha) = 0$

$\therefore a\alpha^4 + b\alpha^3 + c\alpha^2 + d\alpha + f = 0$

$a\alpha^4 + b\alpha^3 + c\alpha^2 + d\alpha = -f$

$\alpha(a\alpha^3 + b\alpha^2 + c\alpha + d) = -f$

since a, b, c, d, f and α are integers, α divides f .

ii) $q(x) = 4x^4 - x^3 + 3x^2 + 2x - 3$

Suppose there is an integer zero, α .

Then α divides -3 , ie. $\alpha = \pm 1$ or ± 3

$q(1) = 4(1)^4 - (1)^3 + 3(1)^2 + 2(1) - 3$
 $= 5$

$q(-1) = 4(-1)^4 - (-1)^3 + 3(-1)^2 + 2(-1) - 3$
 $= 3$

$q(3) = 4(3)^4 - (3)^3 + 3(3)^2 + 2(3) - 3$
 $= 327$

$q(-3) = 4(-3)^4 - (-3)^3 + 3(-3)^2 + 2(-3) - 3$
 $= 369$

None are 0.

$\therefore q(x)$ does not have an integer zero.

1

must mention integers.

1

listing all possible α

1

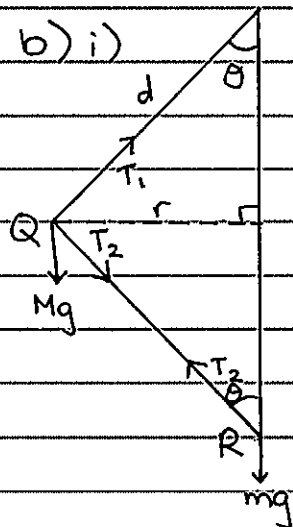
showing none are zero.

1

1

1

b) i)



At R: $T_2 \cos \theta = mg$ ①

At Q:

vertically: $T_1 \cos \theta - T_2 \cos \theta = Mg$ ②

horizontally:

$T_1 \sin \theta + T_2 \sin \theta = M\omega^2 r$

since $\sin \theta = \frac{r}{d}$

$r = d \sin \theta$

$\therefore T_1 \sin \theta + T_2 \sin \theta = M\omega^2 d \sin \theta$

$\therefore T_1 + T_2 = M\omega^2 d$ ③

sub. ① into ②

$T_1 \cos \theta - mg = Mg$

MATHEMATICS Extension 2: Question...14...

Suggested Solutions	Marks	Marker's Comments
$\therefore T_1 \cos \theta = Mg + mg$ $T_1 = (Mg + mg) \sec \theta \quad (4)$ from (1): $T_2 = mg \sec \theta \quad (5)$ Sub (4) and (5) into (3) $\therefore (Mg + mg) \sec \theta + mg \sec \theta = Mw^2 d$ $\therefore (Mg + mg + mg) \sec \theta = Mw^2 d$ $\therefore Mw^2 d = (2mg + Mg) \sec \theta$	1	for working out.
ii) $M = 4m \quad \theta = \frac{\pi}{3}$ $4mw^2 d = (2mg + 4mg) \sec \frac{\pi}{3}$ $4mw^2 d = (6mg)(2)$ $4mw^2 d = 12mg$ $w^2 = \frac{12mg}{4md}$ $= \frac{3g}{d}$ $\therefore w = \sqrt{\frac{3g}{d}}, w > 0$	1	must show substitution
c) i) $ z - 8 = 2 \operatorname{Re}(z - 2)$ let $z = x + iy$ $ x + iy - 8 = 2 \operatorname{Re}(x + iy - 2)$ $\sqrt{(x - 8)^2 + y^2} = 2(x - 2)$ $(x - 8)^2 + y^2 = 4(x - 2)^2$ $x^2 - 16x + 64 + y^2 = 4(x^2 - 4x + 4)$ $x^2 - 16x + 64 + y^2 = 4x^2 - 16x + 16$ $48 = 3x^2 - y^2$ $\therefore \frac{x^2}{16} - \frac{y^2}{48} = 1$	1	
$ z - 8 \geq 0 \quad \therefore 2 \operatorname{Re}(z - 2) \geq 0$ $2(x - 2) \geq 0$ $x - 2 \geq 0$ $\therefore x \geq 2$	1	

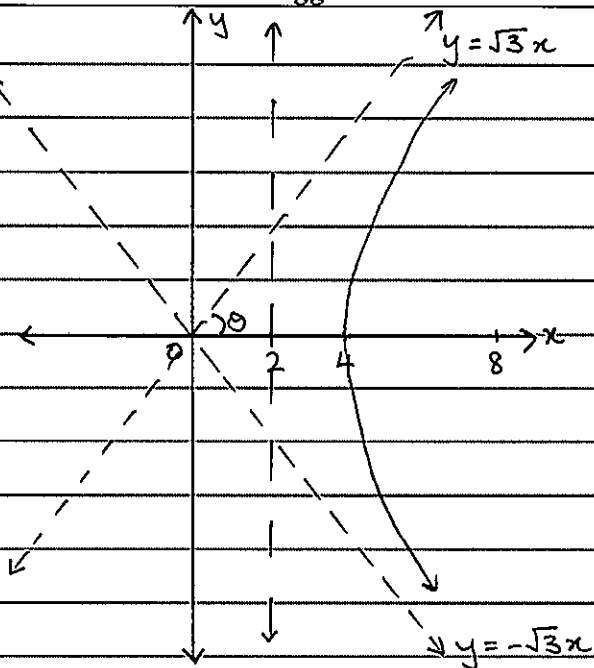
MATHEMATICS Extension 2: Question...14...

Suggested Solutions

Marks

Marker's Comments

ii)



$$a = 4, b = 4\sqrt{3}$$

asymptotes: $y = \pm \frac{b}{a}x$
 $= \pm \sqrt{3}x$

$$y = \sqrt{3}x$$

$$m = \sqrt{3}$$

$$\therefore \tan \theta = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\therefore -\frac{\pi}{3} < \text{Arg } z < \frac{\pi}{3}$$

iii) $|z+8| - |z-8| = 2a$
 $= 8$

1

shape

1

asymptotes

1

1

if no restriction
in part (i)

then $-\frac{\pi}{3} < \text{Arg } z < \frac{\pi}{3}$

and Arg z for
the other branch.
needed for 1 mark.

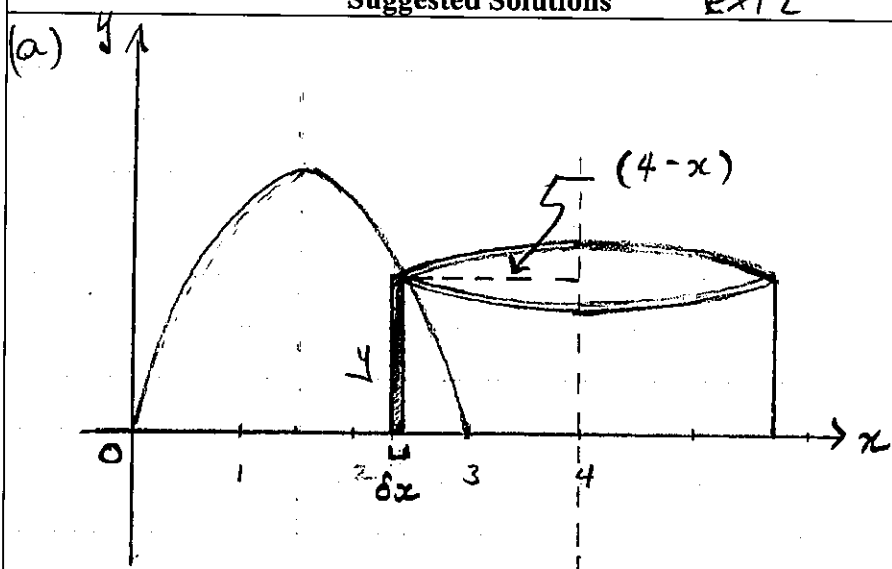
MATHEMATICS: Question 15

Suggested Solutions

EXT 2

Marks

Marker's Comments



Element of volume: $\delta V \approx 2\pi(4-x)y \delta x$ |

So

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 2\pi(4-x)y \delta x$$

$$= \int_0^3 2\pi(4-x)y \, dx$$

$$= 2\pi \int_0^3 (4-x) \cdot x(3-x) \, dx$$

$$= 2\pi \int_0^3 x^3 - 7x^2 + 12x \, dx$$

$$= 2\pi \left[\frac{x^4}{4} - \frac{7x^3}{3} + 6x^2 \right]_0^3$$

$$= 2\pi \left[\frac{81}{4} - 63 + 54 \right]$$

$$= \frac{45\pi}{2} \text{ units}^3.$$

Handled well.

1

1

(b)(i) Let $\alpha^2 = x$ be solution to cubic polynomial (w/ integer coefficients). Then $\alpha = \pm\sqrt{x}$.

1

Reasoned well.

We have, by construction,

$$\alpha^3 - 3\alpha^2 + 5 = 0$$

$$\text{so, } (\pm\sqrt{x})^3 - 3(\pm\sqrt{x})^2 + 5 = 0$$

$$\pm x\sqrt{x} - 3x + 5 = 0$$

$$\Rightarrow (\pm x\sqrt{x})^2 = (3x - 5)^2$$

$$\therefore x^3 = 9x^2 - 30x + 25$$

$$\therefore x^3 - 9x^2 + 30x - 25 = 0$$

1

has solutions $\alpha^2, \beta^2, \gamma^2$

if α, β, γ solutions to

$$x^3 - 3x^2 + 5 = 0$$

//

MATHEMATICS: Question 15

Suggested Solutions

EXT 2

Marks

Marker's Comments

(ii) We have that

$$(\alpha^2 + \beta^2 + \gamma^2)^2 = \alpha^4 + \beta^4 + \gamma^4$$

$$+ 2(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2)$$

From cubic found in part (i),

$$\sum \alpha^2 = 9 \quad \sum \alpha^2\beta^2 = 30,$$

$$\text{so } 9^2 = \sum \alpha^4 + 2(30)$$

i.e.

$$\alpha^4 + \beta^4 + \gamma^4 = 81 - 60$$

$$= 21.$$

Handled well.

1

1

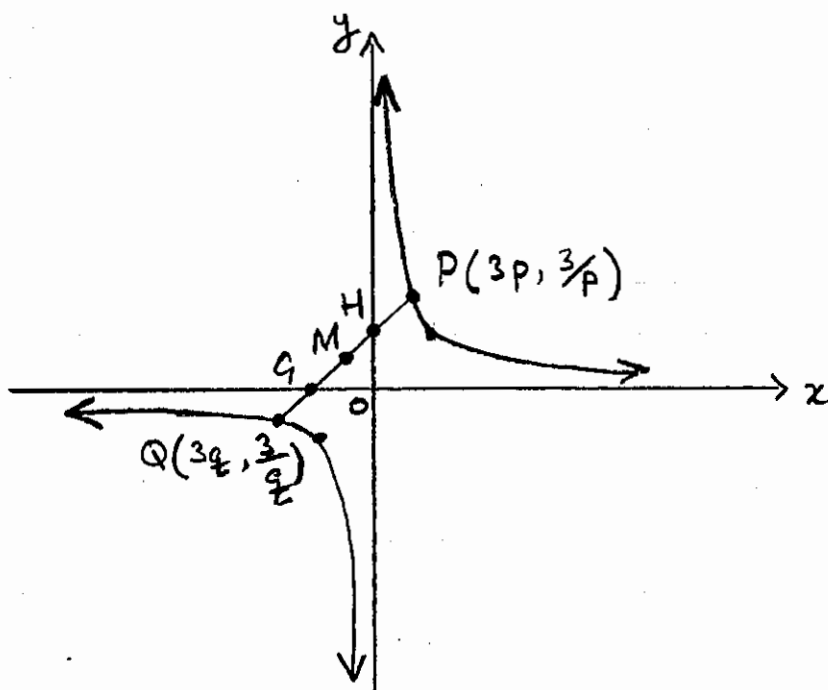
MATHEMATICS: Question 15

Suggested Solutions EXT 2

Marks

Marker's Comments

(c)



$$(i) M = \left(\frac{3(p+q)}{2}, \frac{3(p+q)}{2pq} \right)$$

$$(ii) \text{ Given } H = \left(0, \frac{3(p+q)}{pq} \right)$$

$$G = \left(3(p+q), 0 \right)$$

substituting $x=0$, $y=0$ in eqn of chord respectively, we have

$$PH^2 - QG^2 = \left[(3p)^2 + \left(\frac{3}{p} - \frac{3(p+q)}{pq} \right)^2 \right]$$

$$- \left[(3(p+q) - 3q)^2 + \left(\frac{3}{q} \right)^2 \right]$$

One mark
for some
appropriate
reasoning.

MATHEMATICS: Question 15

Suggested Solutions

EXT2

Marks

Marker's Comments

$$= \left[(3p)^2 + \left(-\frac{3}{q}\right)^2 \right]$$

$$- \left[(3p)^2 + \left(\frac{3}{q}\right)^2 \right]$$

$$= 0$$

$$\text{so } PH^2 = QG^2 \rightarrow PH = QG$$

$$(PH, QG > 0)$$

//

1

One mark
for correct
conclusion.

Could also
have argued.
this geometrically,
but so/n presented
here is that
which majority
of candidature
presented.

MATHEMATICS: Question 15

Suggested Solutions	Marks	Marker's Comments
(iii) Locus: chord passes through $(0, 2)$ always, so, from eqⁿ of chord: $pq(z) + 0 = 3(p+q)$ i.e. $2pq = 3(p+q) \text{ --- ①}$ Let $x = \frac{3}{2}(p+q)$, $y = \frac{3(p+q)}{2pq}$ By ①, $\frac{3(p+q)}{2pq} = 1$, so $y = 1$ for all allowable x. Restrictions: By ①, $q = \frac{3p}{2p-3}$, so $x = \frac{3}{2} \left(p + \frac{3p}{2p-3} \right)$ $\Rightarrow 3p^2 - 2px + 3x = 0$ Real values of p exist here only for $\Delta \geq 0$.	1 1 1	Deriving locus, handled well. First mark for constraint. Next mark for correct locus. Determining restrictions, not handled that well. One mark awarded for making a reasonable start, obtaining one/part of restriction.

MATHEMATICS: Question 15

Suggested Solutions

EXT2

Marks

Marker's Comments

Now,

$$\Delta = (-2x)^2 - 4(3)(3x)$$

$$= 4x^2 - 36x$$

Require $4x^2 - 36x \geq 0$

$$\rightarrow x(x-9) \geq 0$$

$$\rightarrow x \leq 0 \text{ or } x \geq 9$$

But, checking endpoints:

① if $x=0$, then $\frac{3}{2}(p+q)=0$

so $p = -q$.

Also require constraint

$$2pq = 3(p+q)$$

be satisfied; then

$$2p(-p) = 0 \Rightarrow p = 0$$

so $q = 0$ also.

This can't happen since

then P, Q would be

undefined (e.g. $(0, \infty)$)

MATHEMATICS: Question 15

Suggested Solutions

EXT 2

Marks

Marker's Comments

② if $x=9$, then

$$9 = \frac{3}{2}(p+q)$$

so $q = 6 - p$

Checking constraint:

$$2p(6-p) = 3(p+(6-p))$$

$$\rightarrow p^2 - 6p + 9 = 0$$

$$\rightarrow (p-3)^2 = 0$$

so $p=3$, implying $q=3$, too.

But then PQ does not form
a line segment ... and so
no midpoint.

MATHEMATICS: Question 15

Suggested Solutions

EXT 2

Marks

Marker's Comments

Now, if $x > 9$, then

$$\frac{3p^2}{2p-3} > 9$$

$$\Rightarrow 3p^2(2p-3) > 9(2p-3)$$

$$\Rightarrow (2p-3)(p-3)^2 > 9$$

$$\rightarrow p > 3/2$$

But $q = \frac{3p}{2p-3}$ so $q > 0$

if $p > 3/2$... but the

P, Q are on same branch

& $\therefore$ don't generate a chord passing through $(0, 2)$.

So $x > 9$ not allowed

On the other hand, if $x < 0$,

then $(2p-3)(p-3)^2 < 0$

$$\rightarrow p < 3/2$$

MATHEMATICS: Question 15

Suggested Solutions

EXT2

Marks

Marker's Comments

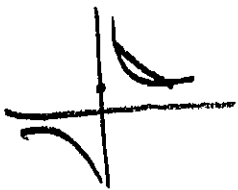
If $0 < p < 3/2$, then

$$\frac{q}{p} = \frac{3p}{2p-3} < 0,$$

so P, Q on separate branches - OK.

If $p < 0$, then $\frac{q}{p} = \frac{3p}{2p-3} > 0$

and, so P, Q again on separate branches.



No chord through $(0, 2)$ if $x > 0$



Chord through $(0, 2)$ if $x < 0$.

Hence, the locus is

$$y = 1 \quad \text{for all real } x < 0.$$

Final mark for correct restriction.

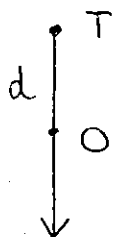
Ext 2 MATHEMATICS: Question 16

Suggested Solutions

Marks

Marker's Comments

a)



$$\ddot{x} = g - kv \quad (\text{given})$$

$$\frac{dv}{dt} = g - kv$$

$$\int_0^v \frac{1}{g - kv} dv = \int_0^t dt$$

$$-\frac{1}{k} [\ln(g - kv)]_0^v = [t]_0^t$$

$$-\frac{1}{k} [\ln|g - kv| - \ln|g - k \cdot 0|] = t$$

$$-\frac{1}{k} [\ln(g - kv) - \ln(g - k \cdot 0)] = t \quad (g > kv)$$

$$-\frac{1}{k} \ln\left(\frac{g - kv}{g - k \cdot 0}\right) = t$$

$$\ln\left(\frac{g - kv}{g - k \cdot 0}\right) = -kt$$

$$\frac{g - kv}{g - k \cdot 0} = e^{-kt}$$

$$g - kv = (g - k \cdot 0) e^{-kt}$$

$$kv = g - (g - k \cdot 0) e^{-kt}$$

$$v = \frac{g}{k} - \frac{(g - k \cdot 0)}{k} e^{-kt}$$

① After Integrating

①

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
ii $\frac{dx}{dt} = \frac{g}{k} - \left(\frac{g-kv}{k}\right) e^{-kt}$ $\int_{-d}^x dx = \int_0^t \left[\frac{g}{k} - \left(\frac{g-kv}{k}\right) e^{-kt} \right] dt$ $\left[x \right]_{-d}^x = \left[\frac{gt}{k} + \left(\frac{g-kv}{k^2}\right) e^{-kt} \right]_0^t$ $x+d = \frac{gt}{k} + \left(\frac{g-kv}{k^2}\right) e^{-kt} - 0 - \left(\frac{g-kv}{k^2}\right)$ $x+d = \frac{gt}{k} + \left(\frac{g-kv}{k^2}\right) e^{-kt} - \left(\frac{g-kv}{k^2}\right)$ $x = \frac{gt}{k} - d + \left(\frac{g-kv}{k^2}\right) (e^{-kt} - 1)$ $x = \frac{gt - kd}{k} + \left(\frac{g-kv}{k^2}\right) (e^{-kt} - 1)$	(1)	For Integrating
iii Substitute $v=0, d=0$ $\therefore v_a = \frac{g}{k} (1 - e^{-kt})$ $x_a = \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1)$	(1) (1)	

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
iv) Collision $\Rightarrow x_p = x_q$ $\therefore \frac{gt - kd}{k} + \left(\frac{g - kv}{k^2} \right) (e^{-kt} - 1) = \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1)$ $-d + \frac{ge^{-kt}}{k^2} - \frac{g}{k^2} - \frac{Ue^{-kt}}{k} + \frac{U}{k} = \frac{ge^{-kt}}{k^2} - \frac{g}{k^2}$ $-d - \frac{Ue^{-kt}}{k} + \frac{U}{k} = 0$ $\frac{Ue^{-kt}}{k} = \frac{U - d}{k}$ $Ue^{-kt} = U - kd$ $e^{-kt} = 1 - \frac{kd}{U}$ $\therefore -kt = \ln \left(1 - \frac{kd}{U} \right)$ $t = \frac{-1}{k} \ln \left(\frac{U - kd}{U} \right)$ $\therefore V_p = \frac{g}{k} - \left(\frac{g - kv}{k} \right) e^{\ln \left(\frac{U - kd}{U} \right)}$ $= \frac{g}{k} - \left(\frac{g - kv}{k} \right) \left(\frac{U - kd}{U} \right)$ $V_a = \frac{g}{k} \left(1 - e^{\ln \left(\frac{U - kd}{U} \right)} \right)$ $= \frac{g}{k} \left(\frac{kd}{U} \right)$ $= \frac{gd}{U}$		① Equating x_p and x_q ①

MATHEMATICS: Question

Suggested Solutions	Marks	Marker's Comments
$\therefore V_p - V_Q = \frac{g}{k} - \left(\frac{g-kU}{k}\right)\left(\frac{U-kd}{U}\right) - \frac{gd}{U}$ $= \left(\frac{g}{k} - \frac{g}{k} + U + \frac{gd}{U} - kd\right) - \frac{gd}{U}$ $= U - kd$ b) : SP is linear in h. Let $SP = mh + b$, when $h=0$, $SP=0$, $\therefore b=0$ $h=H$, $SP=2r$ $\therefore 2r = mH$ $\therefore m = \frac{2r}{H}$ $\therefore SP = \frac{2rh}{H}$		① * Substitution of t correctly - * Showing the subtraction $V_p - V_Q$ * Express independent of Exp & Log
PQ is also linear in h. Let $PQ = nh + p$ when $h=0$, $PQ=4r$ $\therefore p=4r$ when $h=H$, $PQ=2r$ $\therefore 2r = nH + 4r$ $n = \frac{-2r}{H}$ $\therefore PQ = 4r - \frac{2rh}{H}$		① height of rectangle
$\therefore PQ = 4r - \frac{2rh}{H}$	①	base of rectangle

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
$\therefore A_{\text{rectangle}} = \frac{2rh}{H} (4r - \frac{2rh}{H})$		
$A_{\text{semi-ellipse}} = \frac{\pi}{2} (3r - \frac{2rh}{H}) (2r - \frac{rh}{H})$		①
$\therefore A_{\text{cross-section}} = \frac{2rh}{H} (4r - \frac{2rh}{H}) + \frac{\pi}{2} (3r - \frac{2rh}{H}) (2r - \frac{rh}{H})$		correctly substituting into semi-ellipse.
$= \frac{2r^2h}{H^2} (4H - 2h) + \frac{\pi r^2}{2} (3 - \frac{2h}{H}) (2 - \frac{h}{H})$		
$= \frac{4r^2h}{H^2} (2H - h) + \frac{\pi r^2}{2H^2} (3H - 2h)(2H - h)$		
$= \frac{r^2}{2H^2} (8h)(2H - h) + \frac{r^2}{2H^2} (3\pi H - 2h\pi)(2H - h)$		
$= \frac{r^2}{2H^2} (2H - h) (8h + 3\pi H - 2h\pi)$		
$= \frac{r^2}{2H^2} (2H - h) (3\pi H + (8 - 2\pi)h)$		①
$ii) V_{\text{slice}} = \frac{r^2}{2H^2} (2H - h) (3\pi H + (8 - 2\pi)h) \delta h$		
$V_{\text{solid}} = \lim_{\delta h \rightarrow 0} \sum_0^H \frac{r^2}{2H^2} (2H - h) (3\pi H + (8 - 2\pi)h) \delta h$		
$= \frac{r^2}{2H^2} \int_0^H (2H - h) (3\pi H + (8 - 2\pi)h) dh$		
$= \frac{r^2}{2H^2} \int_0^H (6\pi H^2 + 2H(8 - 2\pi)h - 3\pi hH - (8 - 2\pi)h^2) dh$		Expand into an expression that can be integrated

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
$= \frac{r^2}{2H^2} \left[6\pi H^2 h + \frac{H(8-2\pi)h^2}{2} - \frac{3}{2}\pi H h^2 - \frac{(8-2\pi)h^3}{3} \right]_0^H$ $= \frac{r^2}{2H^2} \left[6\pi H^3 + (8-2\pi)H^3 - \frac{3}{2}\pi H^3 - \frac{(8-2\pi)H^3}{3} \right]$ $= \frac{r^2 H}{2} \left[6\pi - 2\pi - \frac{3}{2}\pi + \frac{2\pi}{3} + 8 - \frac{8}{3} \right]$ $= \frac{r^2 H}{2} \left[\frac{19\pi}{6} + \frac{16}{3} \right]$ $= \frac{r^2 H}{12} (19\pi + 32) u^3$		(1)
Some students were awarded 1/2 if they got either the $\frac{19\pi}{12}$ or $\frac{32}{12}$ term in their final answer. (provided it wasn't acquired by accident).		